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Trends and determinants of modern contraceptive use among sexually active adolescent females in Bangladesh: Evidence of four national surveys (2011–2022)

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Abstract

Background Modern contraceptive methods (MCM) reduce the risk of unintended pregnancy and short birth intervals. Despite growing attention to reproductive health, the factors influencing the use of MCM among sexually active adolescent females in Bangladesh remain underexplored. The objectives of this study are to examine trends in the use of MCM and to identify the significant factors associated with the usage of MCM among sexually active adolescent females in Bangladesh.

Methods In this study, we used secondary data from the 2011, 2014, 2017-18, and 2022 rounds of the Bangladesh Demographic and Health Surveys (BDHS) to examine trends in the use of MCM. Additionally, we analyzed data from the selected surveys to identify factors associated with the use of MCM. Male and female sterilization, intrauterine devices (IUDs), injectables, implants, contraceptive pills, male and female condoms, emergency contraception, the standard days method, and the lactational amenorrhea method were considered modern contraceptive methods in the study. A logistic regression model was used to identify the factors influencing the use of MCM.

Results This study revealed that the proportion of the use of MCM among overall surveys was found to be 53.2%, and MCM was more used in urban areas compared to rural areas. It was noted that no consistent pattern was observed in urban, rural, and overall survey. The model identified seven factors associated with the use of MCM among adolescent females: (i) living in the Barisal, Dhaka, Khulna, and Rajshahi divisions; (ii) residing in an urban area; (iii) mass media exposure; (iv) living with one or more children, (v) terminating pregnancy, (vi) visits by a family planning worker; (vii) knowledge on family planning; and (viii) moderate contraceptive-using autonomy.

Conclusion This study highlights key factors influencing the use of MCM among sexually active adolescent females. Addressing these factors-such as strengthening public health sector services, improving access to education, enhancing socioeconomic status, and promoting contraceptive autonomy-is essential to increasing MCM usage among adolescent females.

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Keywords Modern contraceptive methods, Sexually active, Adolescent females, Associated factors, Bangladesh

Introduction

The advent of modern contraceptive (MC) technologies (such as pills, sterilization, intrauterine devices, and emergency contraception) has enabled couples to act on their natural desires for sexual activity with reduced risk of pregnancy at any time chosen [1]. Public health issues in low- and middle-income countries, such as unintended pregnancies, short intervals between pregnancies, and high pregnancy termination rates, are significantly affected by limited access to contraception [2, 3]. The increased use of contraception in developing countries has led to a 40% reduction in unwanted births over the past 20 years, contributing to a decrease in maternal mortality [4]. Access to MC, like pills, IUDs, and implants, is more effective in reducing these risks than relying on traditional methods such as withdrawal or counting days [5, 6]. Still 9% of Bangladeshi women rely on traditional contraceptive methods [7]. The usage of MC improves birth outcomes and reduces complications from closely spaced pregnancies [8, 9]. Moreover, the use of MC methods significantly contributes to lowering maternal mortality by addressing high-risk pregnancies, particularly among adolescent females. This results in a decrease in low birth weight and preterm deliveries [5, 10]. MC-user couples can decide the number of children they want, which helps optimize resources for better maternal and child health [11]. The usage of MC is connected with changes in family size, being told about family planning at health facilities, and being visited by community health workers [12]. Previous studies in Bangladesh have been correlated with enhanced usage of MC, including women's age, occupation, breastfeeding practices, abstaining status of sexual activity in the past year, and women's amenorrhea [13]. Adolescent females constitute approximately 38% of the total female population (83.4 million) in Bangladesh [14]. However, 29.6% of women aged 15–24 years in Bangladesh are still not using modern methods of contraception [15]. One in four (24.65%) adolescent females has ever been pregnant, whereas 3% of them have had a pregnancy loss [7]. Family planning safeguards women from reproductive tract infection, unsafe abortions, and sexually transmitted diseases [4]. To achieve the Sustainable Development Goals (SDGs) 3 target (reducing neonatal mortality to a rate of 12 per 1000 live births), we could improve maternal health by enhancing the usage of MC [16].

The Health Belief Model (HBM) offers a comprehensive framework for understanding the psychological and behavioral mechanisms influencing contraceptive use among sexually active female adolescents [17, 18]. According to the HBM, the use of modern contraceptive methods (MCM) such as pills, injectables, implants, or condoms depends on how young women perceive and evaluate the risks and consequences of unintended pregnancy. Within this framework, family planning education and mass media exposure act as cues to action that raise awareness, correct misinformation, and enhance perceived benefits by emphasizing the safety and effectiveness of contraceptive methods [6, 19]. Education also fosters self-efficacy defined as one's confidence in her ability to make and implement health-related decisions which is crucial for consistent contraceptive use. Furthermore, socioeconomic factors such as wealth status and urban residence may affect perceived barriers by influencing adolescents' access to information and services related to contraception. Therefore, guided by the Health Belief Model, analyzing factors such as education, family planning awareness, media exposure, and socioeconomic status provides a theoretical foundation for identifying actionable determinants that can inform reproductive health interventions among Bangladeshi adolescents.

Many prior studies in Bangladesh have examined the determinants of contraceptive use. However, these studies primarily focused on female adolescents, women of reproductive age, or ever-married women, utilizing BDHS data from 2009 and 2017–18 [15, 20, 21]. A recent study using BDHS 2022 data explored the association between women's empowerment and the use of modern contraceptive methods, finding that women with high empowerment in the social independence domain had 14% lower odds of using modern contraception [22]. Nonetheless, the aforementioned studies did not investigate the determinants of modern contraceptive use among sexually active adolescent females using BDHS 2022 data, leaving a critical gap in understanding this issue in the Bangladeshi context. The objectives of this study are:

(i). To assess changes in the prevalence of MCM use over time; and.
(ii). To identify the socioeconomic, demographic, and autonomy-related factors associated with MCM use among sexually active adolescent females in Bangladesh.

Thus, this study is guided by the following research questions (RQ):

i) RQ1: How to assess changes in the prevalence of MCM use over time?
ii) RQ2: What are the socioeconomic, demographic, and autonomy-related factors are associated with

MCM use among sexually active adolescent females in Bangladesh?

The following null hypotheses are formulated for this study:

i) There is no change in the prevalence of modern contraceptive methods (MCM) use over time.
ii) Socioeconomic, demographic, and autonomy-related factors are not associated with MCM use among sexually active adolescent females in Bangladesh.

Methods

Data source and study population

We extracted data from the Bangladesh Demographic and Health Survey (BDHS) conducted in 2011, 2014, 2017-18, and 2022 to analyze trends and determinants of modern contraceptive methods (MCM) use. These were cross-sectional surveys conducted across Bangladesh. The nationally representative survey, conducted every four years, employs a rigorous two-stage stratified sampling method. During the first stage, enumeration areas (EAs) are selected, followed by the selection of households within each chosen EA in the second stage. This approach ensures the sample is representative of the entire Bangladeshi population. The BDHS targeted households across rural and urban EAs, aiming to complete interviews with women aged 15–49 years. In the present study, we focused on analyzing the women's data file from these comprehensive surveys. BDHS has collected socio-economic, demographic, fertility and mortality, usage of contraceptive, health, and lifestyle information from each selected woman. The details of the sampling technique, data collection procedure, survey design, survey instruments, measuring system, and quality control have been described elsewhere [7].

Inclusion criteria

This study included sexually active Bangladeshi adolescent females aged 15–19 from BDHS datasets (2011: N = 1,749; 2014: N = 1,621; 2017-18: N = 1,597; 2022: N = 1,336), with a final sample of 6,303. Sexual activity was measured as the frequency of sexual intercourse among female adolescents in the last month. The married female adolescents in their reproductive age were asked, "About how many times did you have sex during the last month?" If female adolescents replied that they had at least one sexual encounter in the past month, it was considered sexual activity [23].

Outcome variable

The outcome variable of this study was the usage of MCM. According to the National Institute of Population Research and Training (NIPORT), it was classified into two categories: (i) usage of MCM (coded as 1), and (ii) non-usage of MCM (coded as 0). Male and female sterilization, intrauterine devices (IUDs), injectable, implants, contraceptive pills, male and female condoms, emergency contraception, the standard days method, and the lactational amenorrhea method were considered MCM by BDHS and WHO [7, 24].

Exposure variables

This study selected covariates linked to contraceptive use, including residence (urban/rural), religion, mass media exposure, wealth index, household size, sex of household head, education (woman and husband/partner), employment status, number of children, sex of child, pregnancy termination, family size preference, family planning visits, and husband/partner's occupation. Administrative divisions were Barisal, Chittagong, Dhaka, Khulna, Rajshahi, Rangpur, Sylhet and Mymensingh. Mass media exposure was based on TV, radio, and newspapers, while family planning knowledge included exposure to media or text messages in recent months. Contraceptive autonomy was classified as high (respondent alone), moderate (joint decision), and low (husband/others). We followed some previous studies to select the variables [23, 25–27].

Statistical analysis

Statistical examines were carried out using Stata version 14.2. To accommodate the complex survey design and sampling weights, the 'Svy' command was employed. Descriptive statistics were used to evaluate the occurrence of the use of contraceptives among sexually active adolescent females. Categorical variables were analyzed using Pearson chi-square tests and logistic regression models were employed to compute adjusted odds ratios (ORs) with 95% confidence intervals to examine the relationships between the independent variables and usage of MCM. BDHS collected data from across Bangladesh using a two-stage stratified cluster sampling design, resulting in data with a hierarchical structure. This raised the possibility of a clustering effect on the outcome variables. We assessed the presence of clustering effects using the median odds ratio (MOR), calculated as [MOR = $\exp\{0.6745\sqrt{2}\sqrt{\delta_u^2}\}$=$\exp(0.95\sqrt{\delta_u^2})$, where δ_u^2 represents the cluster-level variance [28].

We found no significant cluster variation in the outcome variables; therefore, a single-level logistic regression model was used for analysis. The multicollinearity problems among the independent variables were checked by the variance inflation factor (VIF). In the analysis, we found the value of VIF lied between 0 and 5 for every selected variable in the logistic model, so there was no evidence of multicollinearity problems among the

independent variables. We performed two-tailed statistical tests, with a significance p-value of less than 0.05.

Results

Table 1 presents the socioeconomic and reproductive status of sexually active adolescent females, based on data combined from four surveys to derive the overall results. Almost 61% of usage of MCM happened in sexually active adolescent females living in the Khulna division of Bangladesh but, its usage was 40% less in the Sylhet division of Bangladesh. The usage of MCM was 59.1% among sexually active adolescent females who live in urban areas. Mass media users of sexually active adolescent females used more than 54% of MCM. Additionally, the usage of MCM among sexually active adolescent females had a higher chance of falling into the following categories: those who were rich wealth index (54.4%); female head of household (44.9%); women with higher education levels (55.8%); those who are working (59.8%); those who have one living child (73.2%); who have a female child (about 55%); pregnancy terminated adolescent females (44.7%); whose house was visited by a family planning worker (64.1%); those who know about family planning (56.3%); and those with high contraceptive-using autonomy (86.1%). Besides, sexually active adolescent girl's husbands/partners who are highly educated (61.5%) and whose husbands want fewer children (57.6%) had used MCM (Table 1).

Trends in usage of MCM

This study examines the percentage of MCM usage among sexually active adolescent females in rural and urban areas of Bangladesh from 2011 to 2022. The proportion of usage of MC among overall surveys was found to be 53.2%, among them 59.1% and 50.4% were in urban and rural areas. Over the four survey years, MC usage was consistently higher in urban compared to rural settings. It was noted that no consistent pattern was observed for the urban, rural, and overall surveys. However, the highest proportion of usage of MC was observed in 2022, and it increased by around 9%, 12% and 8% in the overall sample, urban and rural settings, respectively from 2017 to 18 to 2022 (Fig. 1). Across all survey years, the Z-proportion test revealed a significantly higher prevalence of MCM use among sexually active adolescent females in urban areas than in rural areas ($p < 0.05$). A highly significant difference ($p < 0.01$) in the use of MCM between 2011 and 2022 was observed across urban, rural, and overall samples.It was observed that the higher proportion of usage of MC was found to be in the age group 18–19 for each survey year compared to the age group 15–17 years. However, there was no clear pattern was observed in the usage of MC over four surveys years for both age groups. In particular, the proportion of usage of MC in the age group 18–19 and 15–17 years was increased by approximately 10% and 8% from 2017 to 18 to 2022 surveys respectively (Fig. 2). It was noted that no consistent pattern was observed in the use of pills and injections between 2011 and 2022. However, the use of male condoms and withdrawal methods showed a clear upward trend from 2014 to 2022. In contrast, the use of periodic abstinence declined steadily over the study period, indicating a downward trend from 2011 to 2022 (Table 2).

Factors influencing usage of MCM

Chi-square test showed that administrative divisions, place of residence, sex of household head, women's education level, employment status, number of living children, pregnancy termination, family size concordance, visits by family planning worker, contraceptive-using autonomy and husband/partner's education level were significantly associated with the usage of MC among sexually active adolescent females. Besides, usage of MCM was significantly higher chance among the sexually active adolescent females who had a rich wealth index, who use mass media, and who know family planning (Table 1).

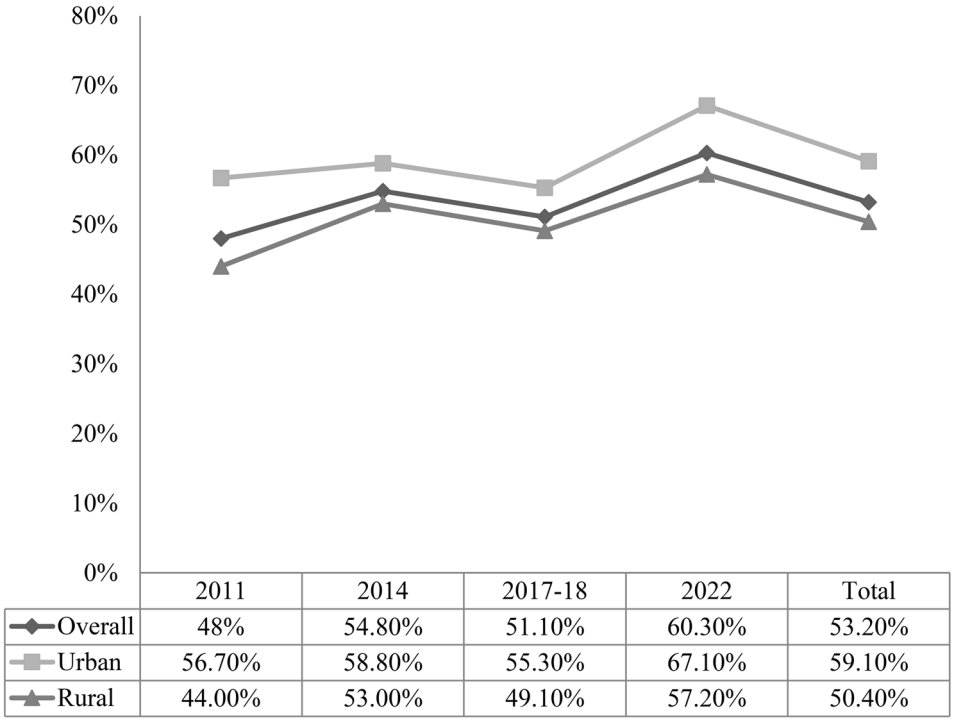
All significant associated factors provided by Chi-square test were considered as independent variables in binary logistic model. The model demonstrated that sexually active female adolescents in Bangladesh from the divisions of Barisal (AOR = 2.43, 95% CI: 1.71–3.45, $p < 0.001$), Dhaka (AOR = 1.69, 95% CI: 1.25–2.29, $p = 0.001$), Khulna (AOR = 2.19, 95% CI: 1.59–3.02, $p < 0.001$), and Rajshahi (AOR = 1.72, 95% CI: 1.25–2.35, $p = 0.001$) had significantly higher odds of using modern contraceptives than those in Chittagong. Urban adolescent girls were 1.42 times more likely to use modern contraceptives (AOR = 1.42, 95% CI = 1.17 to 1.72, $p < 0.001$) compared to rural teenage females. Those exposed to mass media via television, radio, and newspapers had 1.57 times higher odds (AOR = 1.57, 95% CI: 1.30–1.91, $p < 0.001$) of using modern contraceptives compared to those without such exposure. Female adolescents who experienced pregnancy termination had 1.67 times higher odds of using modern contraceptives than those who did not terminate a pregnancy (AOR = 1.67, 95% CI = 1.22 to 2.28, $p = 0.001$). Having a family planning worker visit their home increased the likelihood of contraceptive use by 35% (AOR = 1.35, 95% CI = 1.04 to 1.76, $p = 0.025$). Conversely, those unaware of family planning were less likely to use modern contraceptive methods than those who knew about it (AOR = 0.70, 95% CI: 0.58 to 0.85, $p < 0.001$). A positive link was found between moderate contraceptive-using autonomy and modern contraceptive use, with sexually active adolescents with moderate autonomy showing higher odds of MCM (AOR = 1.33, 95% CI: 1.05 to 1.69, $p = 0.019$). Lastly, the

Table 1 Bivariate association between the usage of modern contraceptives and explanatory variables

Study variables	Total, *N* (%)	Usage of modern contraceptive		*p*-value	Weighted prevalence (95%CI)
		No, *N* (%)	Yes, *N* (%)		
Administrative divisions				< 0.001	
Barisal	717 (11.4)	303 (42.3)	414 (57.7)		53.32 (48.78–57.80)
Chittagong	930 (14.7)	509 (54.7)	421 (45.3)		43.58 (40.65–46.57)
Dhaka	1038 (16.5)	474 (45.7)	564 (54.3)		54.33 (52.97–56.67)
Khulna	902 (14.3)	358 (39.7)	544 (60.3)		58.71 (55.18–62.15)
Rajshahi	888 (14.1)	384 (43.2)	504 (56.8)		55.39 (52.05–58.69)
Rangpur	970 (15.4)	453 (46.7)	517 (53.3)		54.06 (50.76–57.33)
Sylhet	590 (9.4)	355 (60.2)	235 (39.8)		41.42 (36.80-46.19)
Mymensingh	268 (4.2)	116 (43.3)	152 (56.7)		54.65 (48.09–61.05)
Place of residence				**< 0.001**	
Urban	2014 (31.9)	824 (40.9)	1190 (59.1)		58.90 (56.50-61.26)
Rural	4289 (68.1)	2128 (49.6)	2161 (50.4)		49.97 (48.56–51.39)
Religion				**0.647**	
Muslim	5837 (92.6)	2729 (46.7)	3108 (53.3)		52.21 (50.94–53.47)
Non-Muslim	466 (7.4)	223 (47.8)	243 (52.2)		52.78 (48.13–57.38)
Mass media exposure				**0.001**	
No	2222 (35.2)	1106 (49.8)	1116 (50.2)		49.86 (47.77–51.95)
Yes	4081 (64.8)	1846 (45.2)	2235 (54.8)		53.48 (51.97–54.98)
Wealth index				**0.017**	
Poor	2657 (42.1)	1266 (47.6)	1391 (52.4)		51.26 (49.35–53.16)
Middle	317 (5.1)	169 (53.3)	148 (46.7)		44.98 (39.91–50.16)
Rich	3329 (52.8)	1517 (45.6)	1812 (54.4)		53.78 (52.10-55.44)
Household members (in numbers)				**0.091**	
Less than 6	3302 (52.4)	1513 (45.8)	1789 (54.2)		53.51 (51.83–55.19)
6 and above	3001 (47.6)	1439 (47.9)	1562 (52.1)		50.85 (49.08–52.62)
Sex of household head				**< 0.001**	
Male	5776 (91.6)	2662 (46.1)	3114 (53.9)		53.09 (51.81–54.36)
Female	527 (8.4)	290 (55.1)	237 (44.9)		42.99 (38.85–47.23)
Women education level				**< 0.001**	
No education	260 (4.1)	145 (55.8)	115 (44.2)		42.72 (36.92–48.73)
Primary	1496 (23.7)	756 (50.5)	740 (49.5)		47.81 (45.30-50.33)
Secondary	3965 (62.9)	1794 (45.3)	2171 (54.7)		54.17 (52.64–55.69)
Higher	582 (9.3)	257 (44.2)	325 (55.8)		54.77 (50.62–58.85)
Employment status				**< 0.001**	
Not working	5262 (83.5)	2534 (48.2)	2728 (51.8)		50.69 (49.35–52.04)
Working	1041 (16.5)	418 (40.2)	623 (59.8)		59.76 (56.83–62.62)
Number of living children				**< 0.001**	
No child	3206 (50.9)	2114 (65.9)	1092 (34.1)		33.06 (31.46–34.71)
One child	2826 (44.8)	757 (26.8)	2069 (73.2)		71.78 (70.13–73.39)
Two and more child	271 (4.3)	81 (29.9)	190 (70.1)		69.57 (64.01–74.61)
Sex of child				0.983	
Male	1633 (51.6)	749 (45.9)	884 (54.1)		52.92 (50.52–55.31)
Female	1534 (48.4)	703 (45.8)	831 (54.2)		54.19 (51.73–56.64)
Pregnancy termination				**< 0.001**	
No	5813 (92.2)	2681 (46.1)	3132 (53.9)		53.08 (51.73–54.27)
Yes	490 (7.8)	271 (55.3)	219 (44.7)		43.18 (38.88–47.59)
Family size concordance				**< 0.001**	
Couple wants the same	5348 (86.1)	2410 (45.1)	2938 (54.9)		53.68 (52.36-55.00)
Husband wants more	322 (5.2)	154 (47.8)	168 (52.2)		52.68 (47.33–57.98)
Husband wants fewer	243 (3.9)	103 (42.4)	140 (57.6)		56.51 (50.16–62.66)
Don't know	297 (4.8)	196 (65.9)	101 (34.1)		34.86 (29.51–40.63)
Visited by family planning worker				**< 0.001**	

Table 1 (continued)

Study variables	Total, *N* (%)	Usage of modern contraceptive		*p*-value	Weighted prevalence (95%CI)
		No, *N* (%)	Yes, *N* (%)		
No	5334 (84.6)	2604 (48.8)	2730 (51.2)		50.18 (48.85–51.50)
Yes	969 (15.4)	348 (35.9)	621 (64.1)		63.79 (60.73–66.75)
knowledge of family planning				**0.004**	
No	4712 (74.8)	2256 (47.9)	2456 (52.1)		51.46 (50.04–52.87)
Yes	1591 (25.2)	696 (43.7)	895 (56.3)		54.56 (52.14–56.97)
Contraceptive-using autonomy				**< 0.001**	
High	309 (7.5)	43 (13.9)	266 (86.1)		86.04 (81.62–89.53)
Moderate	3297 (80.1)	597 (18.1)	2700 (81.9)		81.10 (79.74–82.39)
Low	508 (12.4)	127 (25.0)	381 (75.0)		74.04 (70.09–77.69)
Husband/partner's education level				**< 0.001**	
No education	756 (12.1)	386 (51.1)	370 (48.9)		49.60 (46.07–53.13)
Primary	2129 (34.1)	1025 (48.1)	1104 (51.9)		49.93 (47.83–52.05)
Secondary	2458 (39.3)	1139 (46.3)	1319 (53.7)		53.13 (51.19–55.06)
Higher	907 (14.5)	349 (38.5)	558 (61.5)		60.73 (57.45–63.91)
Husband/partner's occupation				**0.062**	
Unemployed	128 (2.1)	49 (38.3)	79 (61.7)		63.98 (55.46–71.71)
Farmer/Labour	2952 (47.4)	1411 (47.8)	1541 (52.2)		51.57 (49.80-53.34)
Business	2891 (46.4)	1317 (45.6)	1574 (54.4)		52.94 (51.13–54.74)
Service Holder	261 (4.1)	114 (43.7)	147 (56.3)		56.86 (50.56–62.94)



	2011	2014	2017-18	2022	Total
Overall	48%	54.80%	51.10%	60.30%	53.20%
Urban	56.70%	58.80%	55.30%	67.10%	59.10%
Rural	44.00%	53.00%	49.10%	57.20%	50.40%

Fig. 1 Trends in proportion of usage of modern contraceptive in urban, rural and overall sample over the surveys

odds of using modern contraceptives were 4.94 times higher for adolescents with one living child and 6.59 times higher for those with two or more children, compared to those with no children (AOR = 4.94, 95% CI: 4.09 to 5.97, $p < 0.001$; AOR = 6.59, 95% CI: 3.90 to 11.15, $p < 0.001$) (Table 3).

The area under the receiver operating characteristic (ROC) curve was 0.746, indicating that the model correctly predicted the higher usage of modern contraceptives (MC) in approximately 74.6% of all possible subject pairs (Fig. 3).

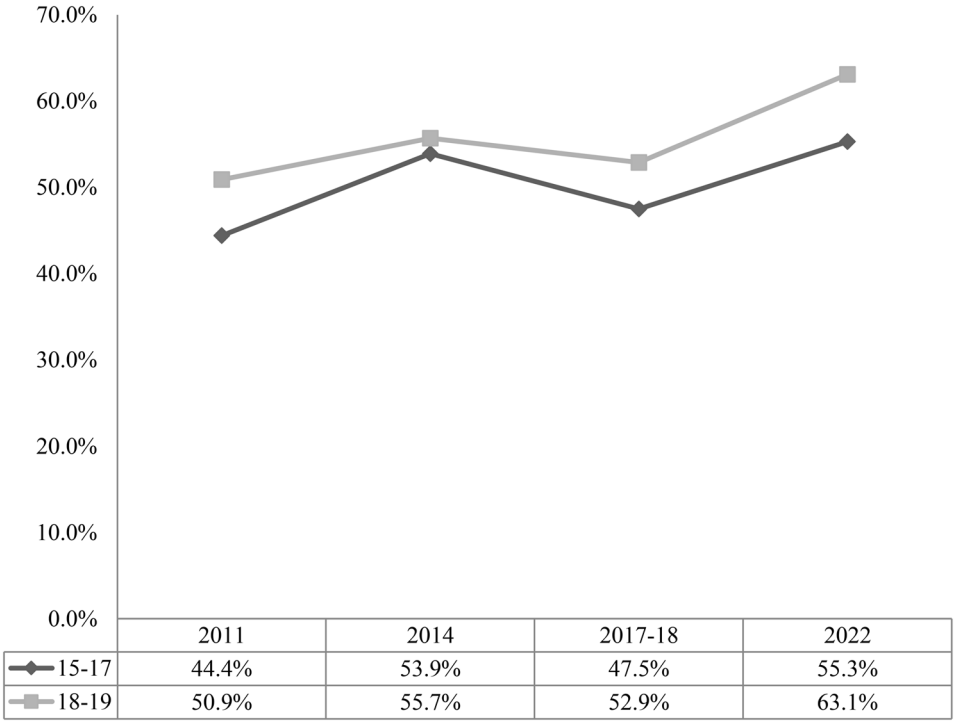


Fig. 2 The differential in modern contraceptive use among sexually active adolescent females by age group over the surveys

Table 2 Modern contraceptive practices among sexually active adolescent females, by survey year

Survey year	**2011**		**2014**		**2018**	**2022**
	***N* (%)**		***N*(%)**		***N*(%)**	***N*(%)**
Method of modern contraceptive						
Pill	524(56.34)		561(57.95)		521(57.57)	429(53.23)
IUD	1 (0.11)		3 (0.31)		0	5 (0.62)
Injections	174 (18.71)		185 (19.11)		113 (12.49)	115 (14.27)
Male condom	124 (13.33)		121(12.50)		149 (16.46)	163 (20.22)
Female sterilization	0 (0.00)		0		2 (0.22)	1 (0.12)
Periodic abstinence	53 (5.70)	5.70	48 (4.96)	4.96	37 (4.09)	30 (3.72)
Withdrawal	37 (3.98)		31 (3.20)	3.20	51 (5.64)	49 (6.08)
Male sterilization	0		1 (0.10)		1 (0.11)	0
Implants/Norplant	17 (1.83)		18 (1.86)		29 (3.20)	9 (1.12)
Lactation amenorrhea	0		0		0	2 (0.25)
Emergency contraception	0		0		2 (0.22)	3 (0.37)
Source of method						
Government	264 (31.43)		249 (28.07)		202 (24.75)	123 (16.97)
Non-Government	39 (4.64)		43 (4.85)		35 (4.29)	37 (5.10)
For-profit outlets and relatives	537 (63.93)		595 (67.08)		579 (70.96)	565 (77.93)

Discussion

This study examines the trends and influential factors of the usage of modern contraceptive methods (MCM) among sexually active adolescent females in Bangladesh.

Trends in MCM

In the present study, we observed that the usage of MCM among adolescent females increased approximately 10% from 2017 to 18 to 2022; however there was no consistent pattern was found from survey year 2011 to 2022. Additionally, higher usage of MCM was found in urban setting as well as in the age group 18–19 compared to the rural and the 15–17 age group. While no clear trend was found across all groups but there was a noticeable increment of MCM usage from the year 2017-18 to 2022 for every group. These shifts were consistent with global findings, as youth-focused sexual and reproductive health (SRH) investments such as school programs,

Table 3 Factors influencing usage of modern contraceptive among sexually active adolescent females in Bangladesh

Study variable	Adjusted aOR		Study variable	Adjusted aOR	
	95% CI for aOR (Lower-Upper)	*p*-value		95% CI for aOR (Lower-Upper)	*p*-value
Administrative divisions			**Pregnancy termination**		
Barisal	2.43 (1.71–3.45)	< 0.001	No	**Reference**	
Chittagong	**Reference**		Yes	1.67 (1.22–2.28)	0.001
Dhaka	1.69 (1.25–2.29)	0.001	**Family size concordance**		
Khulna	2.19 (1.59–3.02)	< 0.001	Husband wants fewer	**Reference**	
Rajshahi	1.72 (1.25–2.35)	0.001	Husband wants more	0.67 (0.54–1.42)	0.185
Rangpur	1.35 (0.99–1.84)	0.055	Couple wants the same	0.87 (0.54–1.42)	0.591
Sylhet	0.95 (0.66–1.34)	0.758	Don't know	0.77 (0.41–1.42)	0.404
Mymensingh	0.95 (0.64–1.41)	0.808	**Visited by family planning worker**		
Place of residence			No	**Reference**	
Urban	1.42 (1.17–1.72)	< 0.001	Yes	1.35 (1.04–1.76)	0.025
Rural	**Reference**		**Knowledge of family planning**		
Mass media exposure			Yes	Reference	
No	**Reference**		No	0.70 (0.58–0.85)	< 0.001
Yes	1.57 (1.30–1.91)	< 0.001	**Contraceptive-using autonomy**		
Wealth index			High	1.47 (0.98–2.22)	0.064
Poor	**Reference**		Moderate	1.33 (1.05–1.69)	0.019
Middle	1.48 (0.87–2.44)	0.133	Low	**Reference**	
Rich	0.89 (0.73–1.09)	0.285	**Husband/partner's education level**		
Sex of household head			No	**Reference**	
Male	**Reference**		Primary	1.07 (0.78–1.46)	0.670
Female	0.83 (0.62–1.11)	0.202	Secondary	1.07 (0.78–1.47)	0.674
Women education level			Higher	1.25 (0.87–1.82)	0.228
No education	**Reference**		**Number of living children**		
Primary	1.07 (0.62–1.83)	0.813	No child	**Reference**	
Secondary	0.95 (0.57–1.61)	0.866	One child	4.94 (4.09–5.97)	< 0.001
Higher	1.37 (0.75–2.51)	0.307	Two and more child	6.59 (3.90-11.15)	< 0.001
Employment status					
Not working	**Reference**				
Working	1.07 (0.85–1.35)	0.565			

peer counseling, and youth-friendly services had been a key driver of this progress. Moreover, increasing education levels and media campaigns on family planning also played a role in shaping more positive attitudes toward MC usage among younger populace [29, 30]. Interestingly, the analysis found no consistent pattern in the use of pills and injectables from 2011 to 2022, which aligns with prior studies showing that adolescent females often use oral pills inconsistently. This may be due to concerns about side effects, as highlighted in previous research [31, 32]. In contrast, male condom usage and withdrawal methods demonstrated an upward trend from 2014 to 2022. The growing popularity of male-controlled methods such as condoms indicated a shift toward shared contraceptive responsibility between partners, particularly in urban areas where awareness and availability were consistently higher. Moreover, studies suggest that adolescents often prefer methods that are easily accessible, do not require prescriptions, and offer privacy [33, 34]. Conversely, the use of periodic abstinence—a traditional method—showed a steady decline from 2011 to 2022. This downward trend endorsed global literature, where reliance on less effective traditional methods decreases with increased access and education about MCM [35]. Reflecting an overall transition toward evidence-based family planning approaches aimed at reducing unintended pregnancies and improving maternal health outcomes.

Factors influencing MCM

Regional differences in adolescent contraceptive use may relate to the availability of MCM, societal norms, and personal preferences [36]. Unequal family planning activities and distribution of resources across administrative divisions exist in Bangladesh [37, 38]. Besides, Bangladeshi women have an incorrect perception about the usage of contraceptives, such as impairing fertility, their use posing serious damage to the user's health, and their being unsafe for prolonged use [39]. Our present study addressed divisional differences as the influential factor

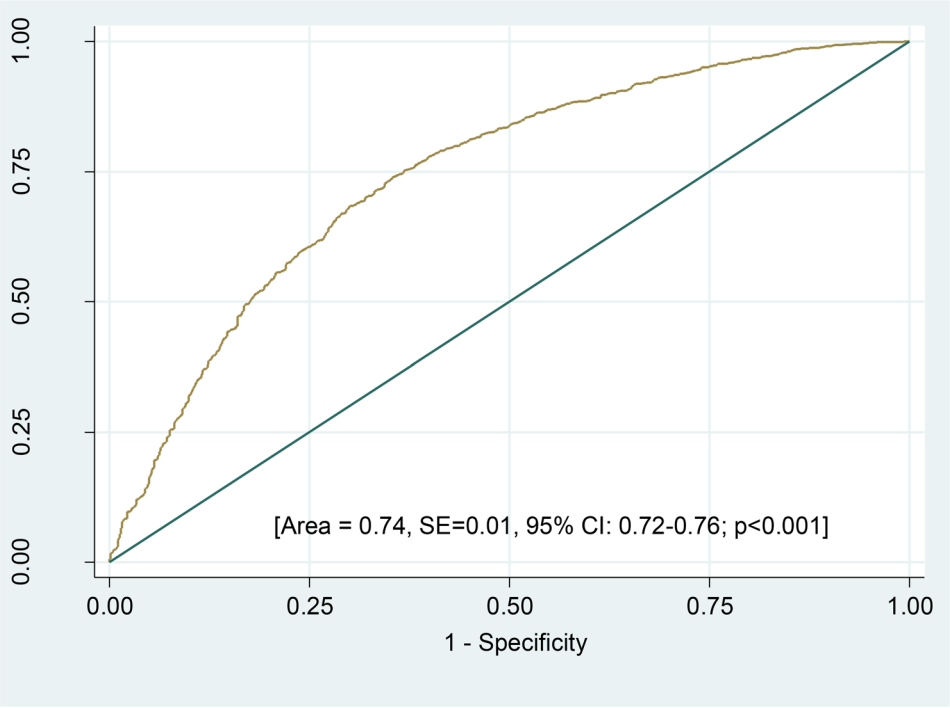


Fig. 3 Roc Curve estimates the goodness of logistic model

on the usage of MCM among sexually active adolescent females in Bangladesh. Adolescent females who resided in the divisions of Barisal, Dhaka, Khulna, and Rajshahi of the country exhibited a higher susceptibility to the usage of MCM compared to their counterparts residing in the Chittagong division. This finding is consistent with previous studies conducted in India [40] and Pakistan [41] Most tribal people are concentrated in the remote, hilly zones of three districts: Bandarban, Khagrachari, and Rangamati, which are part of the Chittagong division [42]. Zafreen et al.'s study shows that 60% of tribal couples in the Chattogram Hill Tract did not use any type of contraceptives [43], which is causative of less usage of modern contraceptives among sexually active adolescent females in the Chittagong division. So, the government should expand family planning workers' visits door-to-door, especially in the Chittagong division, to eliminate the incorrect perceptions about the usage of modern contraceptives. Greater exposure to multiple sources of information, close proximity to healthcare facilities, and easier availability of a broader selection of contraceptive methods in urban areas could have attributed to the more utilization of MCM usage [44, 45]. Urban areas offer better access to education [46], whereas educated women have a higher likelihood of modern contraceptive use [47]. Educated women have a better chance of being in a better position to negotiate the usage of MCM with their partners [48]. Our investigation revealed that adolescents from urban areas were more likely to use contraception compared to those in rural areas, consistent with prior studies which also found adolescent females residing in urban areas had a higher tendency to utilize MCM [25, 40]. Furthermore, our study found a significant influence of mass media (television viewing, radio listening, and newspaper reading) and the use of modern contraception. This finding is consistent with previous research conducted in India [49], Pakistan [50], and Nepal [51], which suggests that the use of modern contraceptive methods (MCM) increases significantly with greater media access. This pattern aligns with the Diffusion of Innovations Theory [52], which explains how exposure to mass media facilitates the adoption of new health behaviors such as contraceptive use. From a policy perspective, media-health partnerships could further enhance the impact of strengthening family planning in Bangladesh. The Number of living children has a significant impact on adolescents' MCM usage. Our study indicated that adolescents with one, two, or more living children had a higher susceptibility to utilize MCM. Prior research also validated this finding when the number of living children becomes higher, women are more willing to depend on MC techniques [53]. Once adolescent females engage in sexual activity, they may be required to demonstrate their ability to conceive, regardless of the reproductive health consequences. The desire to have a first child is the biggest factor in MCM utilization [54]. Non-utilization of contraceptives was frequently linked to unintended pregnancy, which frequently resulted in induced

pregnancy termination, with a considerable proportion being unsafe attempts which greatly adds to the burden of illness and death [55, 56]. This finding aligns with the Theory of Planned Behavior, suggesting that adolescents' contraceptive decisions are shaped by perceived behavioral control and fertility-related attitudes. Strengthening postpartum and adolescent-friendly reproductive health services could help promote informed fertility decisions among young mothers [57]. Our study affirmed a significant association between pregnancy termination and the utilization of MC among sexually active adolescent females, and revealed that prior experience of pregnancy termination influenced the greater usages of modern contraception, which aligns with the previous studies [58]. Another important factor related to the use of modern contraceptives by female adolescents identified in this study is a visit from a family planning worker. The likelihood of using modern contraceptives among female adolescents who were visited by a family planning worker increased compared to those who were not visited. Our findings are consistent with a Pakistani study [47]. Knowledge of family planning crucially impacts the utilization of contraception by increasing an individual's understanding, shaping their attitude, and potentially leading them to make a well-informed choice to use the services [59]. This finding is explained by the Theory of Planned Behavior, which posits that more knowledge and previous reproductive experience strengthen behavioral intentions towards preventive practices such as uptake of modern contraceptives [57]. Our study depicted those adolescents who knew about family planning used MCM more than those who did not. The existing body of literature also suggests that the availability of family planning information for adolescent females through different methods leads to a higher utilization of MCM usage [60, 61]. Contraceptive- using autonomy was also considered an influential factor in the usage of MCM among sexually active adolescent females in Bangladesh. This trend reflected findings from a study conducted in rural Malawi [62], demonstrated that adolescents who had contraceptive-using autonomy significantly increased the likelihood of utilizing MCM. This emphasizes the need for improving adolescent-friendly reproductive health services and augmenting community programs facilitating girls' ability to make informed decisions on contraception. Practical suggestions include offering training for healthcare professionals in adolescent counseling and reducing sociocultural constraints on decision-making independence [63, 64].

Strengths and Limitations

The key strength of this study is that it employed four BDHS datasets from 2011 to 2022, which is a robust large sample from a population-based survey. This study considered the only sexually active adolescent girls in Bangladesh for the use of MCM. Our findings are more reliable with the use of a nationally validated survey. Advanced statistical methods employed in this paper include the use of logistic regression models for identifying hierarchical data structures within the BDHS datasets to determine the factors that influence the usage of MCM. Besides, we use BDHS data from 2011 to 2022 to determine the trend line of modern contraceptive usage among adolescent girls. Despite its strengths, the study has some limitations. First, BDHS relies on self-report data that is subject to recall bias and social desirability bias, which may distort the accuracy of self-reported usage of modern contraceptives. Additionally, we considered only currently sexually active married women aged 15–19 years for this study. Second, due to its cross-sectional design, the study cannot make causal inferences between the usage of modern contraceptives among sexually active female adolescents and their associated factors. Third, some possibly relevant variables that might influence the usage of modern contraceptives, which did not, measured in this study, such as household income, cultural norms, attitudes toward contraception at an individual level of dynamics between partners, and access to healthcare, etc.

Conclusions and recommendations

Between 2011 and 2022, modern contraceptive use among sexually active female adolescents in Bangladesh remained higher in urban areas than in rural ones. The pill was the most commonly used method, mainly accessed from private outlets and relatives. However, about 40% of adolescents still did not use any modern contraceptive method (MCM) in 2022. Key determinants of MCM use included administrative division, media exposure, number of children, pregnancy termination, family planning visits, and knowledge of contraception. Moderate contraceptive autonomy also increased MCM use.

To improve uptake, family planning programs should be context-specific and target lagging divisions, particularly Chittagong. Awareness campaigns through peer educators and youth-friendly services can reduce misconceptions and promote informed choices. Strengthening communication by family planning workers and engaging parents in reproductive health education are vital. Future longitudinal or mixed-method studies should assess the impact of school-based reproductive health policies on adolescent contraceptive behavior.

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ChatGPT-4.0 was used to enhance the readability and language quality of the manuscript.

Author contributions

MAH designed the study and had full access to the data. MAH validated and performed the statistical analysis. MAH, MYA, BRS and MM worked on the original draft of the manuscript, with MAH contributing to the drafting process. MAH, MYA, BRS , ASMAM, MGH and MM reviewed the manuscript critically. ASMAM and MGH supervised the entire study. All authors read and approved the final version of the manuscript.

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Data availability

The BDHS datasets and relevant materials are freely available at https://dhsprogram.com/data/dataset/Bangladesh_Standard-DHS_2022.cfm?flag=0 https://dhsprogram.com/data/dataset/Bangladesh_Standard-DHS_2017-18.cfm?flag=0 https:// dhsprogram.com/data/dataset/Bangladesh_Standard-DHS_2014.cfm?flag=0 [https://dhsprogram.com/data/dataset/Bangladesh_Standard-DHS_2011.cfm?flag=0](https:/dhsprogram.com/data/dataset/Bangladesh_Standard-DHS_2011.cfm?flag=0).

Declarations

Ethics approval and consent to participate

The present study was based on secondary analysis of publicly available data from the Demographic and Health Surveys (DHS). Ethical approval for the original data collection was obtained by the DHS Program; therefore, no additional ethical approval was required for this study. As the authors did not directly engage with human participants, adherence to specific ethical guidelines, such as the Declaration of Helsinki, is not applicable to this secondary data analysis. The Bangladesh Demographic and Health Survey (BDHS) obtained written informed consent from all participants prior to data collection.

Competing interests

The authors declare no competing interests.

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